HALLIBURTON

Apr 27 2023 | Logged in as : Goytia, Renzo Nicolas NicolÃ¡s

# Introduction

# Baroid Core Homework 15

Assessment Feedback

Congratulations, you passed.

Total score: 138 out of 144, 96%

Question Feedback

## Question

1 of 51

Organize in the right order the steps to perform the calculation of materials required to build a Non Aqueous Fluid

| | |
|---|---|
| Calculate density of the fluid phase | 2 |
| Calculate the volume and quantity of the weighting agent | 3 |
| Calculate volume and ratio mass of all additives | 1 |
| Calculate the volume of base fluid | 4 |
| Calculate the volume of water and pounds/sacks of salt | 5 |

5 out of 5

## Question

2 of 51

What is the function of BARABLOK™ in a NAF?

- ○ Viscosifier
- ○ Weighting agent
- ○ Rheology modifier
- ○ Emulsifier
- ○ Wetting agent
- ○ Thinner
- ○ Suspension agent
- ● Filtration control agent
- ○ Alkalinity source

1 out of 1

# Question

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What is the function of GELTONE® II in a NAF?

- Weighting agent
- Emulsifier
- Viscosifier
- Filtration control agent
- Thinner
- Wetting agent
- Rheology modifier
- Suspension agent
- Alkalinity source

1 out of 1

# Question

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What is the function of RM-63™ in a NAF?

- Thinner
- Alkalinity source
- Weighting agent
- Filtration control agent
- Suspension agent
- Viscosifier
- Emulsifier
- Wetting agent
- Rheology modifier

1 out of 1

# Question

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What is the function of INVERMUL® in a NAF?

- Alkalinity source
- Filtration control agent
- Weighting agent
- Viscosifier

- Suspension agent
- Thinner
- Rheology modifier
- Emulsifier
- Wetting agent

1 out of 1

## Question

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What is the function of DURATONE® HT in a NAF?

- Rheology modifier
- Alkalinity source
- Emulsifier
- Weighting agent
- Wetting agent
- Viscosifier
- Filtration control agent
- Suspension agent
- Thinner

1 out of 1

## Question

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What is the function of EZ MUL® NT in a NAF?

- Filtration control agent
- Viscosifier
- Emulsifier
- Weighting agent
- Alkalinity source
- Rheology modifier
- Suspension agent
- Thinner
- Wetting agent

1 out of 1

## Question

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What is the function of SUSPENTONE™ in a NAF?

- ○ Wetting agent
- ○ Thinner
- ○ Viscosifier
- ○ Filtration control agent
- ○ Rheology modifier
- ◉ Suspension agent
- ○ Weighting agent
- ○ Alkalinity source
- ○ Emulsifier

1 out of 1

## Question

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What is the function of DRILTREAT® in a NAF?

- ○ Suspension agent
- ◉ Wetting agent
- ○ Viscosifier
- ○ Rheology modifier
- ○ Emulsifier
- ○ Thinner
- ○ Alkalinity source
- ○ Filtration control agent
- ○ Weighting agent

1 out of 1

## Question

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What is the function of OMC® 42™ in a NAF?

- ○ Wetting agent
- ○ Filtration control agent
- ○ Viscosifier
- ○ Suspension agent
- ○ Emulsifier
- ○ Weighting agent
- ○ Rheology modifier
- ◉ Thinner
- ○ Alkalinity source

1 out of 1

# Question

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What is the function of RHEMOD™ L in a NAF?

- ○ Weighting agent
- ◉ Viscosifier
- ○ Rheology modifier
- ○ Suspension agent
- ○ Alkalinity source
- ○ Filtration control agent
- ○ Thinner
- ○ Wetting agent
- ○ Emulsifier

1 out of 1

# Question

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Why is lime added into Non Aqueous fluids?

- [x] To increase the alkalinity
- [ ] To decrease the alkalinity
- [ ] To activate some viscosifiers
- [ ] To increase the salinity
- [ ] To increase the content of calcium
- [x] To activate some emulsifiers

2 out of 2

# Question

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Why is calcium chloride added into Non Aqueous fluids?

- [ ] Lower the fluid loss into the formation
- [x] To increase the salinity
- [ ] Activate some thinners
- [x] Control osmotic exchange with the formation
- [ ] To increase the alkalinity

2 out of 2

## Question

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Which of the following options are true about the properties of premium diesel

- SG= 0.80; density = 6.7ppg
- SG= 0.84; density = 7ppg
- SG= 0.86; density = 7.18ppg
- SG= 0.82; density = 6.84ppg

1 out of 1

## Question

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Which of the following products is recommended to be added into NAF, when extra barite is added to a NAF?

- OMC® 42™
- Lime
- DRILTREAT®
- BARABLOK™
- All of the above
- None of the above

1 out of 1

## Question

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List in chronological order how the products/materials should be mixed to build a NAF

| | |
|---|---|
| 5th | Viscosifiers |
| 7th | Wetting agent |
| 2nd | Emulsifiers |
| 1st | Base Fluid |
| 6th | Filtration control agents |
| 3rd | Alkalinity source |
| 8th | Weighting agents |
| 4th | Brine |

8 out of 8

## Question

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Select all the parameters that are needed to calculate the materials required to build a NAF

- [x] Volume of the fluid
- [x] NAP/Water ratio
- [ ] Flow line temperature
- [x] Water phase salinity
- [ ] Mixing time of the fluid
- [ ] All of the above

3 out of 3

# Question

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How many barrels of oil (SG = 0.79), barrels of water, sacks of $CaCl_2$ salt (80lb/sack), and sacks of barite (100lb/sack) are needed to build 2,300 bbl of 13.6 ppg invert emulsion fluid with a NAP/water ratio of 80/20 and 260,000ppm salinity?

How many barrels of NAP are needed? 1424.32

3 out of 3

# Question

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For NAF Build 1, how many barrels of water are needed?

Barrels of water = 323.32

3 out of 3

# Question

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For NAF Build 1, how many sacks of barite (100lb/sack) are needed?

Sacks of barite (100lb/sack) = 7649

3 out of 3

# Question

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For NAF Build 1, how many sacks of calcium chloride (80lb/sack) are needed?

Sacks of calcium chloride (80lb/sack) = 519

3 out of 3

## Question

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Build 4,000 bbl of a 13.2 invert emulsion fluid with an NAP/water ratio of 70/30, a WPS of 260,000ppm. Calculate how many tons of barite (2000lb/ton), sacks of CaCl2 (80lb/sack), barrels of water, and barrels of oil (7.0 ppg) will be needed to build the drilling fluid

How many barrels of NAP are needed? = 2261.76

3 out of 3

## Question

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For NAF Build 2, how many barrels of water are needed?

Barrels of water = 880.14

3 out of 3

## Question

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For NAF Build 2, how many tons of barite (2000lb/ton) are needed?

Tons of barite (2000lb/ton) = 565.95

3 out of 3

## Question

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For NAF Build 2, how many sacks of calcium chloride (80lb/sack) are needed?

Sacks of calcium chloride (80lb/sack) = 1411

3 out of 3

## Question

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Build 1,750 bbl of an invert emulsion fluid with a density of 13.4ppg, NAP/water ratio of 95/05, a WPS of 340,000 ppm and the non aqueous phase of 7.09ppg, Calculate how much oil, water, Barite (100lb/sack), and CaCl2 salt (80lb/sack) will be needed

How many barrels of NAP are needed? = 1297.99

0 out of 3

NAP = 1296.58bbl

## Question

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For NAF build 3, how many barrels of water are needed?

Barrels of water = 58.96

3 out of 3

## Question

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For NAF build 3, how many sacks of barite (100lb/sack) are needed?

Sacks of barite (100lb/sack) = 5649

0 out of 3

Barite = 5670sxs

## Question

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For NAF build 3, how many sacks of calcium chloride (80lb/sack) are needed?

Sacks of calcium chloride (80lb/sack) = 140

3 out of 3

## Question

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Calculate the oil, water, calcium chloride (80lb/sack), and barite (100lb/sack) required to mix 1,050 bbl of a 13.2 ppg invert emulsion drilling fluid with a WPS of 260,000ppm and NAP/water ratio of 85/15. The density of the oil is 7.0ppg.

How many barrels of NAP are needed? = 707.84

3 out of 3

## Question

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For NAF build 4, how many barrels of water are needed?

Barrels of water = 113.42

3 out of 3

# Question

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For NAF build 4, how many sacks of barite (100lb/sack) are needed?

3198

3 out of 3

# Question

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For NAF build 4, how many sacks of calcium chloride (80lb/sack) are needed?

Sacks of calcium chloride (80lb/sack) = 182

3 out of 3

# Question

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Calculate all the materials required to build 1,000bbl of an IEF system of 11.3ppg with a NAP/Water ratio of 75/25, WPS of 210,000ppm.
The density of the NAP base fluid is 7ppg. Weighting material is barite (100lb/sack).

The formulation to use is:

| Product | Package size | Specific Gravity | Concentration (lb/bbl) |
|---|---|---|---|
| INVERMUL NT | 55 gal/drum | 0.94 | 5.0 |
| Lime | 50 lb/sack | 2.40 | 4.0 |
| GELTONE V | 50 lb/sack | 1.70 | 5.0 |
| DURATONE HT | 50 lb/sack | 1.80 | 3.0 |
| DRILTREAT | 55 gal/drum | 1.00 | 1.5 |

How many barrels of NAP are needed? = 628.14

3 out of 3

# Question

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For NAF build 5, how many barrels of water are needed?

Barrels of water = 195.14

3 out of 3

## Question

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For NAF build 5, how many sacks of barite (100lb/sack) are needed?

Sacks of barite (100lb/sack) = 1842

3 out of 3

## Question

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For NAF build 5, how many sacks of calcium chloride (80lb/sack) are needed?

Sacks of calcium chloride (80lb/sack) = 236

3 out of 3

## Question

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For NAF build 5, how many drums (55 gal/drum) of INVERMUL are needed?

Drums of INVERMUL = 12

3 out of 3

## Question

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For NAF build 5, how many sacks of lime (50lb/sack) are needed?

Sacks of lime = 80

3 out of 3

## Question

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For NAF build 5, how many sacks of GELTONE V are needed?

Sacks of GELTONE V = 100

3 out of 3

## Question

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For NAF build 5, how many sacks of DURATONE HT (50lb/sack) are needed?

Sacks of DURATONE HT = 60

3 out of 3

## Question

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For NAF build 5, how many drums (55 gal/drum) of DRILTREAT are needed?

Drums of DRILTREAT = 4

3 out of 3

## Question

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Given the following data from a NAF fluid:
Retort oil: 54% (SG 0.84) and Retort water: 17%
Chloride: 38,000mg/L Calcium: 15,000mg/L MW: 14.5ppg.

Calculate the Water Phase salinity in parts per million

WPS (ppm) = 262300

5 out of 5

## Question

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Using the following data, complete the Water Phase Salinity analysis for a sample of BaraXcel™ NAF IEF:

Fluid Density [ppg] = 11.8

Calcium Content [mg/l] = 16,000

Chlorides [mg/l] = 22,000

Water from retort = 9%

Oil from retort = 72%

Specific Gravity of Oil = 0.84

What is the Water phase salinity in parts per million? = 276700

3 out of 3

# Question

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What is the specific gravity of the brine?

$SG_{brine}$ = 1.25431

3 out of 3

# Question

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What is the ASG of the fluid system?

ASG = 3.785

3 out of 3

# Question

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For WPS analysis 3, what is the percentage and concentration of Low gravity solids?

%LGS= 4.69 $LGS_{ppb}$ = 42.74

6 out of 6

%LGS=4.69$

LGSppb=42.74ppb

# Question

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For WPS analysis 3, what is the percentage and concentration of High gravity solids?

%HGS = 13.39 HGSppb = 197.11

6 out of 6

%HGS=13.39%

HGSppb=197.11ppb

# Question

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Given the following information about a NAF containing a calcium chloride-based brine and the **information from the $CaCl_2$ brine table**:

Retort oil: 52% (SG 0.82)    Water: 16%

WPS: 230,000ppm    MW: 15.9ppg

Calculate the information requested below.

What is the oil ratio? 76 What is the water ratio? 24 What is the SG of the brine? 1.21

5 out of 5
OWR= 76/24
SGbrine=1.21

# Question

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What is the percentage volume fraction of the brine?

*(Refer to API 13B-2, Section 12.6.1.1, Formula # 56 )*

%Volume of the brine (%) = 17.17

3 out of 3

# Question

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What is the ASG of the fluid system?

ASG (3 decimals) = 4.122

3 out of 3